

Executive Summary

Loan Approval Prediction Using Machine Learning

Objective

The objective of this project is to build a supervised machine learning model that predicts loan approval decisions based on borrower characteristics. The project emphasizes data preprocessing, handling class imbalance, comparing multiple classification models, and evaluating their predictive performance using standard metrics.

Methodology

The dataset was preprocessed by handling missing values, encoding categorical variables, and scaling numerical features. Since the target variable exhibited class imbalance, Synthetic Minority Oversampling Technique (SMOTE) was applied to create a balanced training dataset.

Two classification algorithms were implemented:

- Logistic Regression
- Random Forest Classifier

Model performance was evaluated using Precision, Recall, F1-score, and ROC-AUC metrics.

Key Findings

- SMOTE improved the model's ability to identify minority class observations.
- Random Forest captured nonlinear relationships more effectively than Logistic Regression.
- Credit history and applicant income emerged as influential factors in loan approval decisions.
- ROC-AUC analysis indicated good discrimination capability for both models.

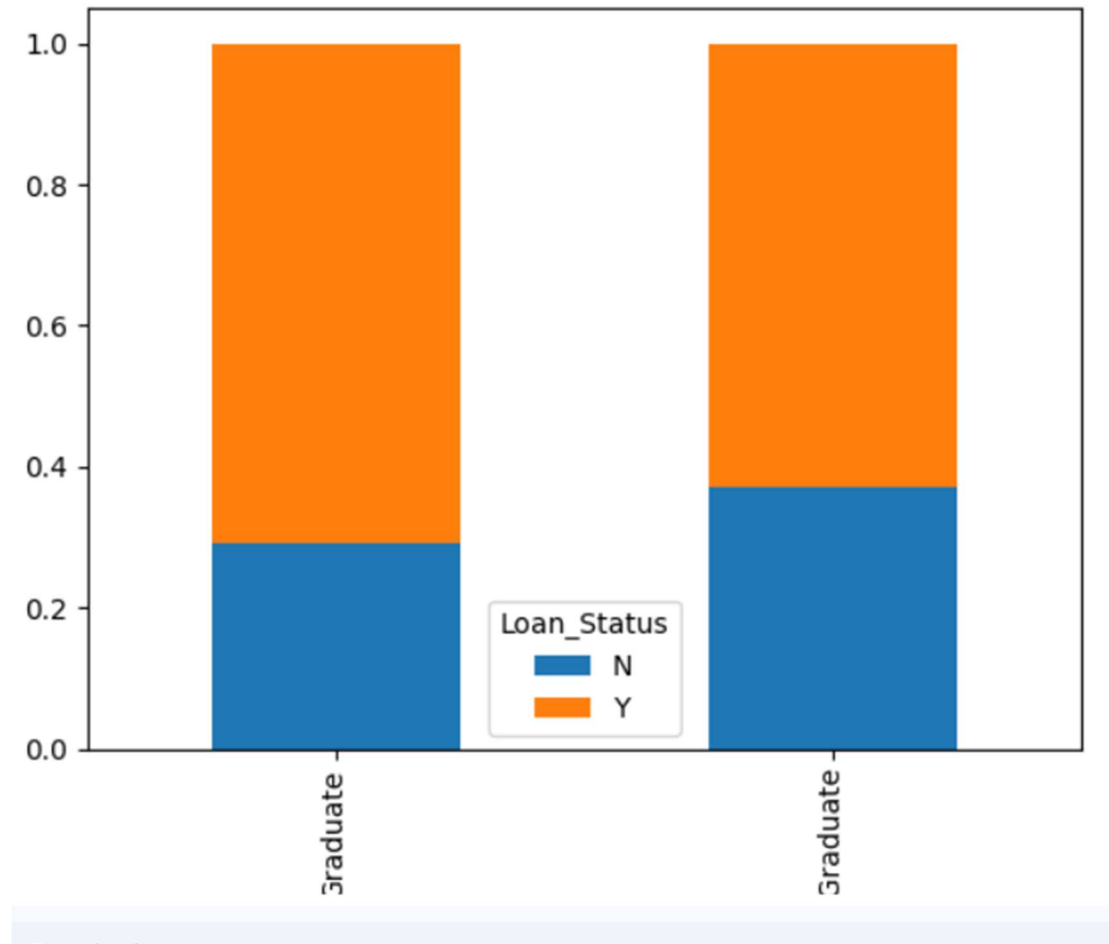
Business Interpretation

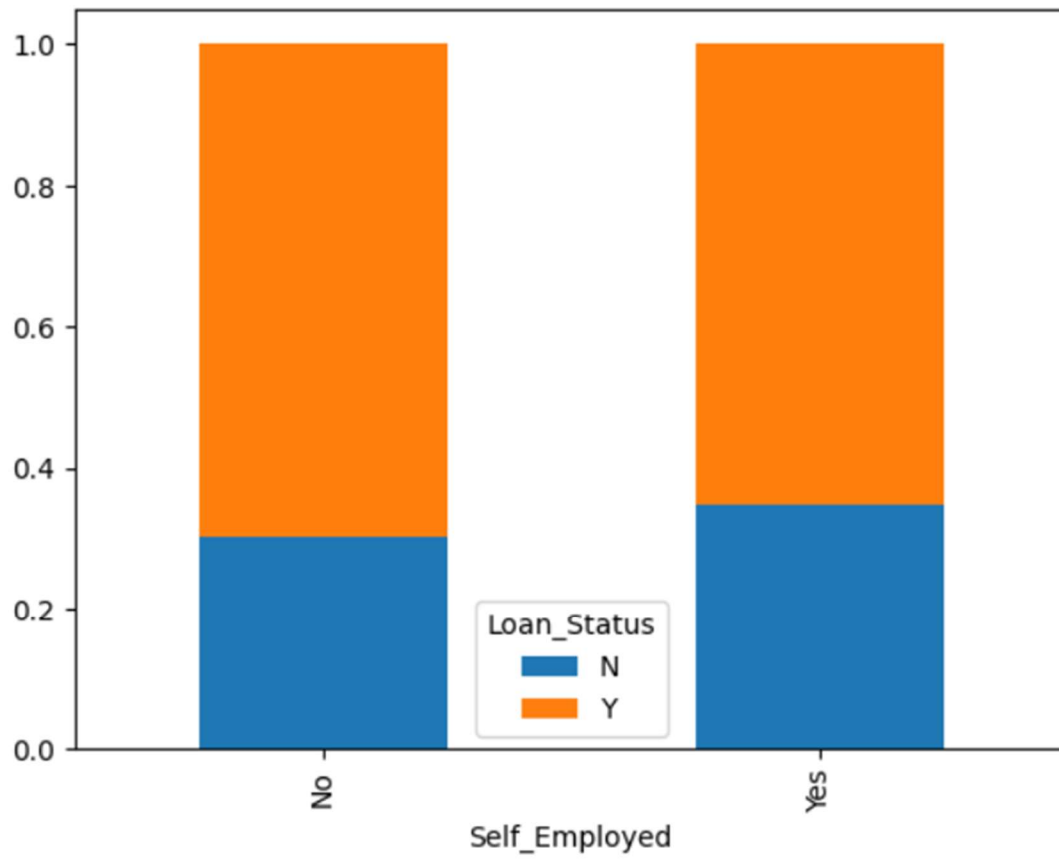
The predictive model can assist financial institutions in automating preliminary loan screening while reducing manual workload. It enables faster decision-making, improves consistency, and supports risk management by identifying applicants with higher probabilities of default or rejection.

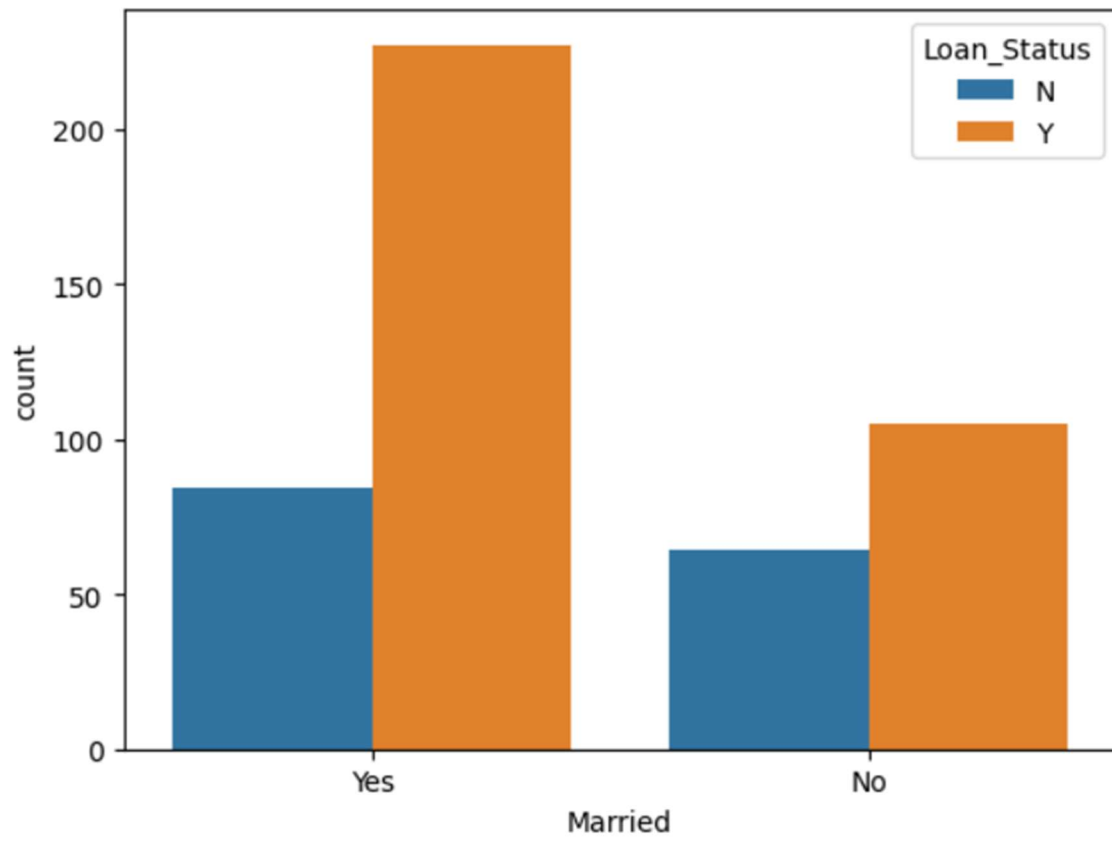
Recommendation

Deploy the Random Forest model with an optimized probability threshold determined through Precision-Recall or ROC analysis to balance business objectives between minimizing loan defaults and maximizing approval rates.

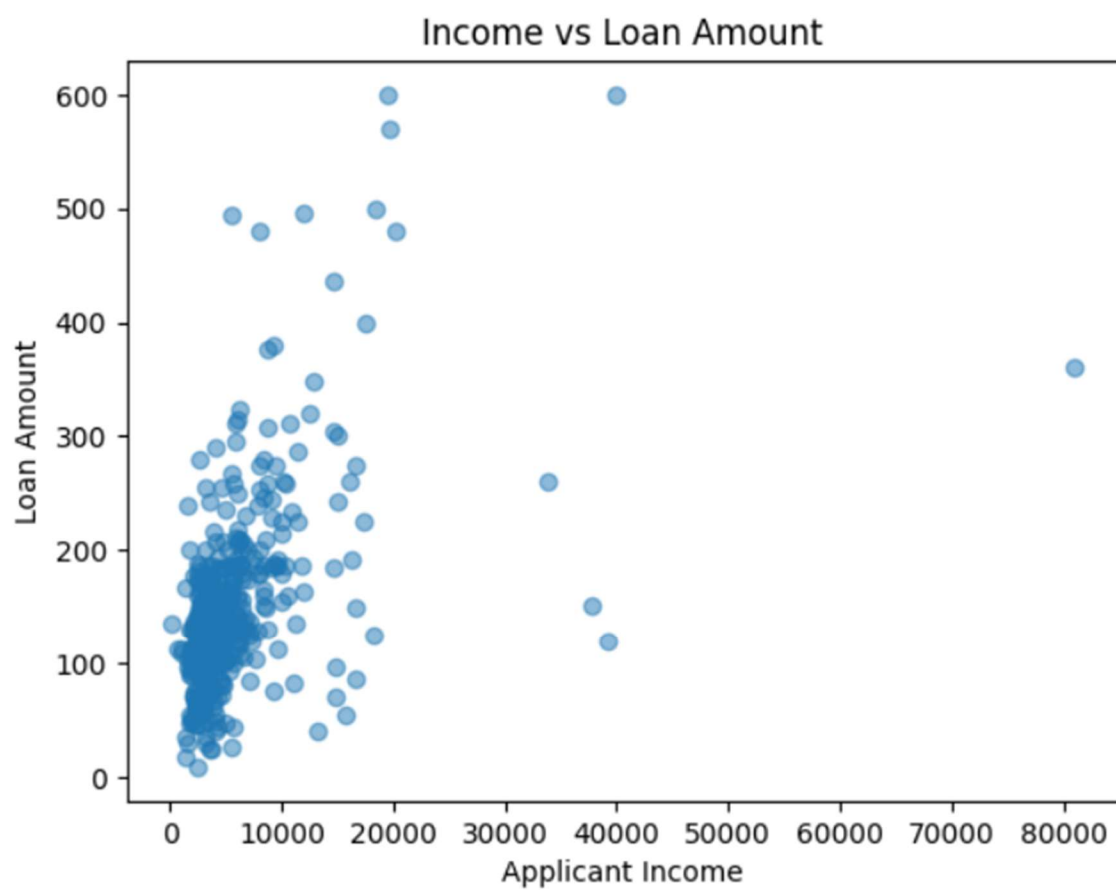
Screenshots







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Credit History vs Loan Approval

